Week 3: Operations Research Models

CCS4004 - Optimization Techniques

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Today's Journey

1. What are OR Models?
2. Structure of OR Models
3. Classic OR Model Families
4. Building OR Models
5. Let's Practice!
6. OR Models in Industry

Quick Question

How does Tesla decide what to produce, where to ship, and how much to stock?

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Answer: Operations Research Models

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From Real World to Models

Operations Research (OR)

A scientific approach to decision making that uses **models** to represent complex real systems.

Why Models?

- Real systems are too complex to test directly
- Models let us experiment *on paper* or on a computer
- We can search for optimal or good decisions

Real System $\Rightarrow$ **Model** $\Rightarrow$ **Solution** $\Rightarrow$ **Implementation**

Types of Operations Research Models

By Representation

- Physical models
- Symbolic / diagrammatic models
- Mathematical models

By Certainty

- Deterministic models
- Stochastic (probabilistic) models

By Time

- Static
- Dynamic

By Structure

- Linear / Nonlinear
- Integer / Mixed-integer
- Network, Queuing, Simulation

Choosing the right model type is the first key step.

Physical vs Mathematical Models

Physical Models

- Scaled-down or simplified replicas
- Example: Factory layout mock-up
- Good for visualization
- Hard to optimize numerically

Use When

You need intuition and communication with non-technical audiences.

Mathematical Models

- Variables, constraints, objective
- Example: Linear programming model
- Can be solved by algorithms
- Easily changed and tested

Use When

You want quantitative decisions and optimization.

Deterministic vs Stochastic Models

Deterministic Models

- All parameters known with certainty
- No randomness in inputs
- Same input $\Rightarrow$ same output

Examples

- Classical linear programming
- Basic transportation problem
- Production planning with fixed demand

Stochastic Models

- Some parameters are random
- Use probabilities / distributions
- Same input $\Rightarrow$ different outcomes possible

Examples

- Queuing models (random arrivals)
- Inventory with random demand
- Reliability / failure models

Static vs Dynamic Models

Static Models

- One-shot decision
- Single time period
- No explicit time dynamics

Examples

- Single-period product mix
- One-time investment allocation
- One-day staff scheduling

Dynamic Models

- Decisions over multiple periods
- State evolves over time
- Often uses ODEs or difference equations

Examples

- Multi-period inventory
- Equipment replacement
- Dynamic pricing

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General Mathematical Model

Objective Function

Maximize or Minimize $f(x_1, x_2, \ldots, x_n)$

Constraints

$$g_1(x_1, \ldots, x_n) \leq b_1$$
$$g_2(x_1, \ldots, x_n) = b_2$$
$$\vdots$$

x_j in allowed domain (continuous, integer, binary)

Same structure for LP, IP, NLP, network, and many other OR models.

Decision Variables, Parameters, Constraints

Decision Variables

- What we can choose or control
- Example: number of units to produce, trucks to send, workers to schedule

Parameters

- Known input values (data)
- Example: unit profit, resource capacity, demand, travel cost

Constraints

- Represent physical or logical limits
- Example: capacity, demand satisfaction, logical relationships

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Standard OR Model Families

- **Allocation models** (resource allocation)
- **Transportation and assignment models**
- **Network flow models** (shortest path, max-flow)
- **Inventory models**
- **Queuing models**
- **Simulation models**
- **Game theory models**
- **Dynamic programming models**

Many real problems can be mapped to one of these prototypes.

Allocation Models

Goal: Allocate scarce resources among competing activities.

Example: Product Mix

$$\begin{aligned} \max \quad & \sum_{j=1}^{n} p_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^{n} a_{ij} x_j \leq b_i, \quad i = 1, \ldots, m \\ & x_j \geq 0, \quad j = 1, \ldots, n \end{aligned}$$

Interpretation

- x_j: quantity of product j to produce
- p_j: profit per unit
- a_{ij}: amount of resource i used by product j
- b_i: available amount of resource i

Transportation Models

Goal: Ship goods from sources to destinations at minimum cost.

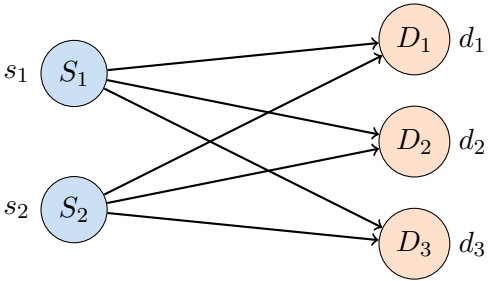
Decision Variables

- x_{ij} = units shipped from source i to destination j

Model

$$
\begin{aligned}
\min \quad & \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij} \\
\text{s.t.} \quad & \sum_{j=1}^{n} x_{ij} = s_i, \quad i = 1, \ldots, m \quad \text{(supply)} \\
& \sum_{i=1}^{m} x_{ij} = d_j, \quad j = 1, \ldots, n \quad \text{(demand)} \\
& x_{ij} \geq 0
\end{aligned}
$$

Transportation Network Diagram



Each path has a cost c_{ij}, and the model chooses optimal flows x_{ij}.

Assignment Models

Goal: Assign agents to tasks with minimum cost or maximum efficiency.

Decision Variables

- $x_{ij} = 1$ if worker i is assigned to job j, else 0

Model

$$
\begin{aligned}
\min \quad & \sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij} x_{ij} \\
\text{s.t.} \quad & \sum_{j=1}^{n} x_{ij} = 1, \quad i = 1, \ldots, n \quad \text{(each worker gets 1 job)} \\
& \sum_{i=1}^{n} x_{ij} = 1, \quad j = 1, \ldots, n \quad \text{(each job gets 1 worker)} \\
& x_{ij} \in \{0, 1\}
\end{aligned}
$$

Network Flow Models

Idea: Represent system as a graph (nodes + arcs) with flows.

Common Problems

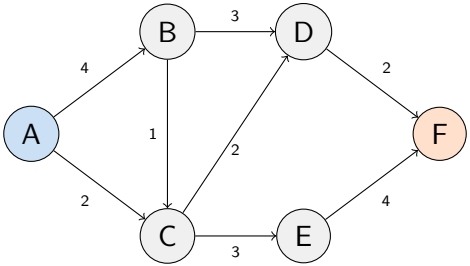
- Shortest path
- Maximum flow
- Minimum spanning tree
- Minimum cost flow

Applications

Transport, telecommunications, computer networks, supply chains, project planning.

Shortest Path Problem

Goal: Find minimum distance (or cost) path from source to destination.



Real Uses: GPS navigation, routing internet packets, robot path planning.

Inventory Models

Goal: Decide how much and when to order to balance cost and service.

Key Costs

- Ordering cost (setup cost)
- Holding (carrying) cost
- Shortage or backorder cost

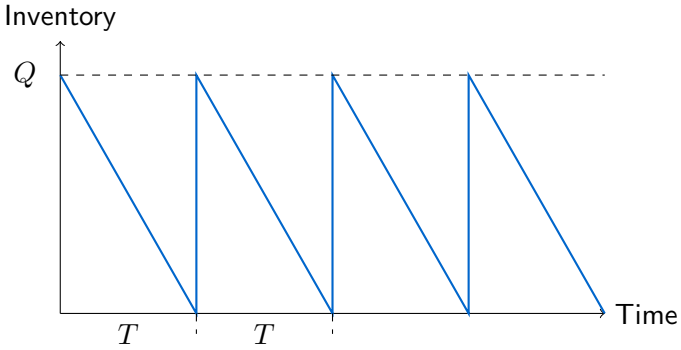
Basic EOQ Formula (no shortages)

$$Q^* = \sqrt{\frac{2DS}{H}}$$

where:

- D = annual demand
- S = ordering cost per order
- H = holding cost per unit per year

Inventory Level Over Time



Trade-off: Large Q (fewer orders) vs high holding cost.

Queuing Models

Goal: Analyze waiting lines to balance service quality and cost.

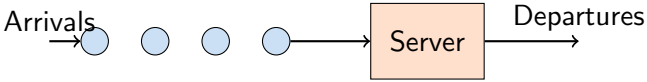
Elements of a Queue

- Arrival process (e.g., Poisson)
- Service process (e.g., exponential)
- Number of servers
- Queue discipline (FIFO, priority)

Applications

- Call centers, hospitals, banks
- Computer servers, networks
- Airport security, supermarkets

Single-Server Queue Diagram



Performance Measures: Average wait, queue length, server utilization.

Simulation Models

When to Use Simulation

- System too complex for analytical formulas
- Many random elements and interactions
- You want to test *what-if* scenarios

Examples

- Simulating a whole hospital
- Simulating an airport or seaport
- Simulating internet traffic

Key Idea

Build a computer model and **sample** system behavior many times.

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Steps in Building an OR Model

1. **Define the problem clearly**
2. **Collect and analyze data**
3. **Formulate the model**
4. **Verify and validate** the model
5. **Solve** the model using appropriate methods
6. **Interpret** and test solutions
7. **Implement** and monitor performance

Model is only useful if it reflects reality well enough.

Example: Warehouse Location Problem

Scenario

- A company serves customers in several cities
- It can open warehouses in candidate locations
- Each warehouse has a fixed opening cost
- Shipping cost depends on distance

Goal

- Decide which warehouses to open
- Decide how to assign customers to warehouses
- Minimize total cost (fixed + shipping)

Warehouse Location: Model

Decision Variables

- $y_j = 1$ if warehouse j is opened, 0 otherwise
- $x_{ij} = 1$ if customer i is served by warehouse j, 0 otherwise

Model

$$
\begin{aligned}
\min \quad & \sum_j F_j y_j + \sum_i \sum_j c_{ij} x_{ij} \\
\text{s.t.} \quad & \sum_j x_{ij} = 1, \quad \forall i \quad \text{(each customer assigned)} \\
& x_{ij} \le y_j, \quad \forall i, j \quad \text{(cannot use closed warehouse)} \\
& y_j \in \{0, 1\}, \; x_{ij} \in \{0, 1\}
\end{aligned}
$$

This is a mixed-integer programming model.

Model Assumptions Matter

Common Assumptions

- Demand is known and constant
- Costs are linear and additive
- No capacity limits (or simple capacities)
- Travel times ignore traffic or variability

Dangers

- Over-simplification
- Wrong conclusions in practice

Always question: Are these assumptions reasonable for this problem?

Model Validation

Validation Questions

- Does the model reproduce known past behavior?
- Are outputs reasonable for extreme input values?
- Do domain experts agree with the structure?

Simple Checks

- Units are consistent
- Constraints reflect real limitations
- Optimal solution is interpretable

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Challenge: Restaurant Staffing

Scenario

- A restaurant is open from 12:00 to 22:00
- Customer demand varies by hour
- Each worker shift is 4 hours
- You want enough staff but not too many

Your Task

1. Define decision variables
2. Propose an objective function
3. Suggest key constraints

Restaurant Staffing: Possible Model

Decision Variables

- x_t = number of workers starting a shift at time slot t

Objective

$$\min \sum_t c x_t \quad \text{(minimize total staffing cost)}$$

Constraints

- For each hour h:

  workers on duty $\geq$ demand at hour h
- Shifts last exactly 4 hours
- $x_t \geq 0$ and integer

This becomes an integer programming model.

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Case Study: Airlines

Major OR Problems

- Fleet assignment to routes
- Crew scheduling and rostering
- Ticket pricing and overbooking

Model Types

- Large-scale integer programming
- Network flow models
- Stochastic models for uncertainty

Impact

Even small improvements can save millions of dollars per year.

Case Study: E-commerce (e.g., Amazon)

OR Challenges

- Where to place warehouses and hubs
- How to route delivery trucks daily
- How much inventory to keep at each location

Model Types Used

- Location and allocation models
- Vehicle routing and network models
- Inventory and queuing models